

JOSH T. MARTINO

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EDUCATION

James Madison University | B.S. Information Technology, Minor in Robotics **May 2027**
Leadership: Co-Founder, Madison Tech Society | Web Development Committee Head, Madison Artificial Intelligence Club

EXPERIENCE

- Project Manager & Undergraduate Research Assistant - JACart** | CCI Scholars, James Madison University **Jun 2025 - Present**
Harrisonburg, VA
- Develop and integrate autonomous-vehicle software across ROS 2 sensor, localization, and AI systems; built a real-time anomaly-detection pipeline using vehicle telemetry and camera context.
 - Led technical planning and cross-team coordination for JACart research, translating faculty objectives into development milestones, testing priorities, and integration work.
 - Improved system reliability through ROS 2 debugging and integration across LiDAR, ZED cameras, localization, Dockerized services, and networked vehicle components.
 - Helped migrate localization from NDT to MOLA ICP and integrated localization-health diagnostics into the autonomy and anomaly-detection stack.
- Undergraduate Researcher - HGN Impairment Detection** | James Madison University **Aug 2026 - Present**
Harrisonburg, VA
- Research computer-vision and eye-tracking methods for detecting Horizontal Gaze Nystagmus and other ocular indicators associated with driver impairment.
 - Review impairment-detection literature and candidate eye-movement features to guide future model development and data-collection decisions.
- Teaching Assistant - Advanced Python Programming** | James Madison University **Jun 2024 - May 2025**
Harrisonburg, VA
- Supported 50+ students through debugging, tutoring, and live coding; taught labs covering binary search trees, graph algorithms, and Dijkstra's algorithm.

PROJECTS

- DayLy - Full-Stack Scheduling & Task Management System** | *React, Supabase, FullCalendar*
- Built a full-stack planner with month/week views, recurring RRULE events, category and priority filtering, timed/all-day tasks, and real-time task persistence.
 - Designed reusable React components for task creation, detail views, and a Notion-style navigation experience while coordinating feature and architecture decisions.
- SentinelNet - Security Event Triage Dashboard** | *Python, AWS CDK, Lambda, SQS, DynamoDB, Wazuh*
- Engineered a Wazuh-to-SQS-to-Lambda ingestion pipeline and DynamoDB backend; validated end-to-end processing against hundreds of security alerts.
 - Defined cloud infrastructure with AWS CDK and implemented security controls including KMS encryption, Cognito group-based access, and API authorization.
- JMULifts - Campus Rideshare Application** | *Swift, Firebase, HCI*
- Developed an iOS rideshare prototype for the JMU community with Firebase email-link authentication and university-domain access controls.
 - Iterated on user flows and interface designs through HCI deliverables focused on usability, trust, and low-friction ride coordination.
- Type 1 Diabetes Adverse Event Prediction - Digital Health Capstone** | *Python, Deep Learning, Time-Series*
- Developing a deep-learning pipeline to predict adverse events from physiological time-series data, emphasizing useful prediction lead time in addition to model accuracy.
 - Defining data-cleaning, event-labeling, and baseline-evaluation workflows, with a later phase planned for real-time wearable data and an application interface.
- GPT-2 Reimplementation & Training Optimization** | *PyTorch*
- Reimplemented GPT-2 with causal self-attention, transformer blocks, weight tying, and autoregressive generation.
 - Optimized training with mixed precision, gradient accumulation, learning-rate scheduling, gradient clipping, and efficient data loading.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL
AI / ML: PyTorch, scikit-learn, Pandas, NumPy, Transformers, LLM/VLM Systems, NLP
Cloud / Backend: AWS, AWS CDK, Lambda, SQS, DynamoDB, Docker, REST APIs
Systems / Web: ROS 2, Linux, Ollama, React, Vite, Git, GitHub, Computer Networking

LEADERSHIP & ACTIVITIES

- CGI Consulting Challenge - Participant** | *Jan 2025*
- Teamed with 3 peers to develop a data-backed strategy for a live client and presented recommendations to CGI consultants, earning top-quartile feedback.